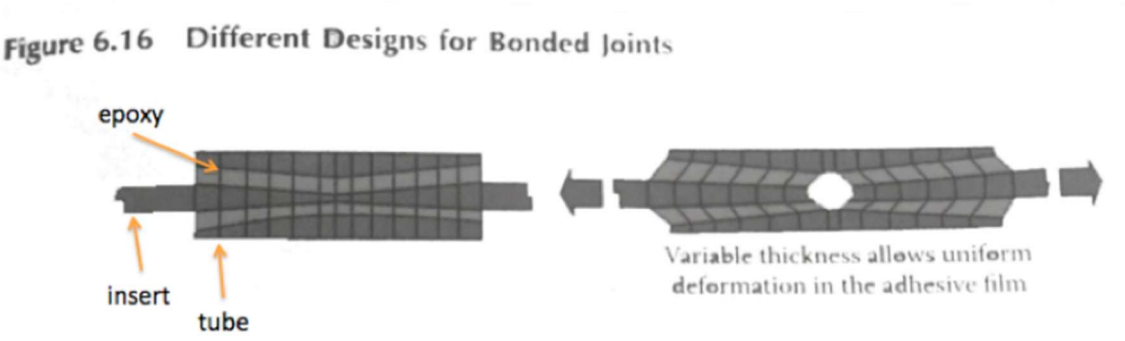
Types of Carbon Fiber Tubing:



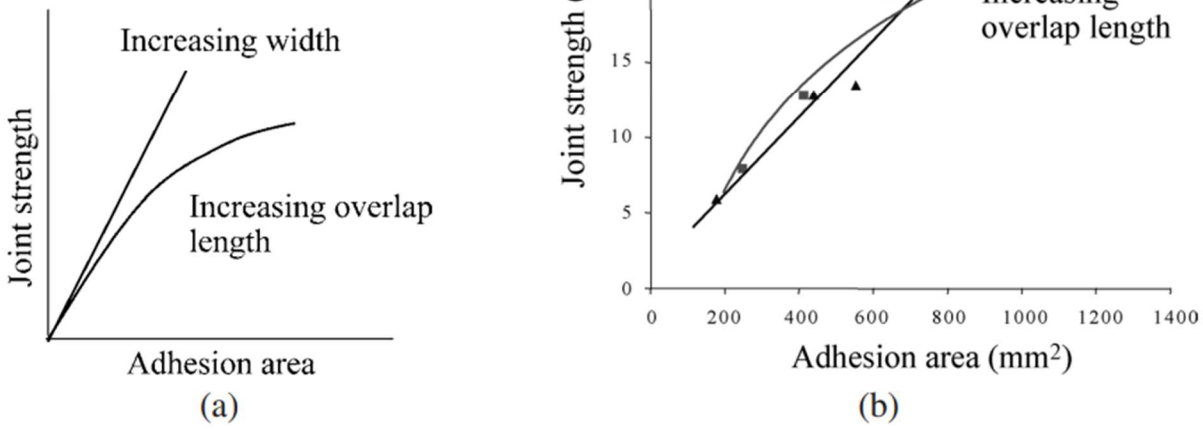
- Suspension links are usually treated as two-force members (compression-tension), **roll-wrapped** carbon fiber tubing would help to treat them as such because of its higher torsional strength, axial strength, and lateral strength, though they are more expensive.
- When carbon fiber tubes fail, most of the time it is under compression

~~**Aluminum insert to Carbon Fiber tube bonding:**~~



Will look at tapering after more imminent deliverables - not much research performed on this "tapering" method specifically, but I can determine area of each material as a function of x and use this assuming generalized Hooke's Law.

Joint Strength vs Adhesion Area:



Increasing bonding circumference and overlap length helps with joint strength, but it is asymptotic with overlap length - this is known as the Volkersen theory. Therefore, future testing experiments should be more concerned with varying increasing widths, and this varies the joint strength the most.

Assuming both adherends are rigid and made from the same material, the adhesive deforms only under shear stress:

$$\tau = \frac{P}{bl}$$

Where τ is the shear stress, P is the load, b is the joint width (circumference since cylindrical), and l is the overlap length. From this research, the need to calculate bending moment for a pin-and-collar assembly, which is what we use, is unnecessary. There are also shear strength limits found, accounting for the difference in joint area:

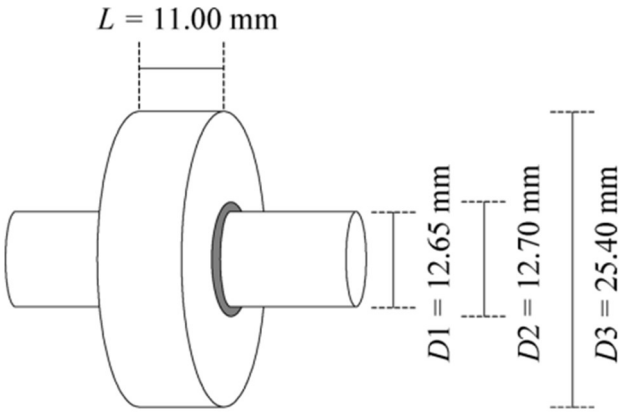
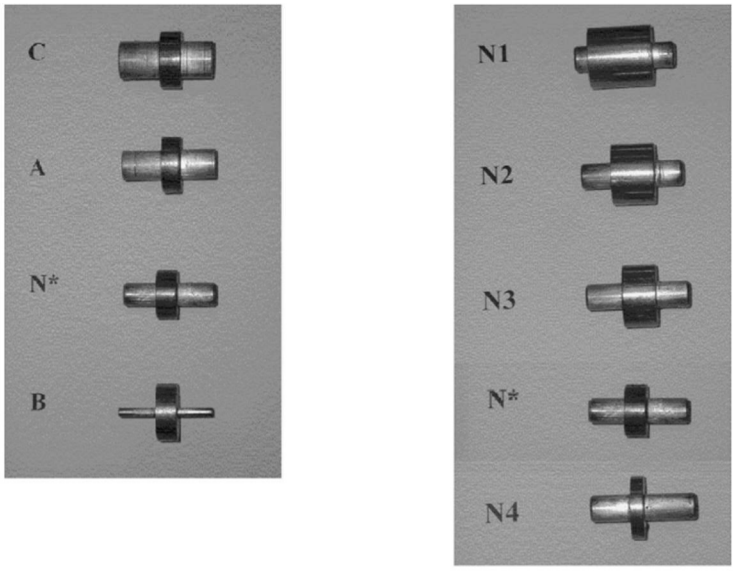


Table 1.
Nominal dimensions of pin-and-collar specimens

Configuration	$D2$ (mm)	$D1$ (mm)	L (mm)	Joint area (mm^2)
C	20.00	19.86	11.00	691.15
A	16.00	15.96	11.00	552.92
B	5.15	4.95	11.00	177.15
N* (ISO 10123)	12.70	12.65	11.00	438.88
N1	12.70	12.65	31.32	1249.61
N2	12.70	12.65	21.10	841.85
N3	12.70	12.65	16.10	642.36
N4	12.70	12.65	6.20	247.37

Configurations are shown in Fig. 5.



Group I Group II

Table 2.
Shear strength values (in kN) obtained in the pin-and-collar assembly tests

	Configuration	Loctite 603	Loctite 638
Varying joint width	C	18.09	19.86
	A	12.42	13.47
	N*	10.70	12.80
	B	4.29	5.90
Varying overlap length	N1	20.38	21.12
	N2	15.06	24.44
	N3	13.13	19.59
	N*	10.70	12.80
	N4	7.44	7.91

The above research suggests that one of the most important factors for these setups is the carbon-fiber tubing ID, as we will see with following equations of shear strength along the overlap length.

Adhesive Bonding

*Good resource is Shigley's Mechanical Eng. Design 11th Edition, Section 9-9, pg. 508

Shigley:

- Good design practice is to have adhesive carry the load in shear, rather than in tension

- Volkersen had developed a closed-form stress model for how shear varied along the length of bonding. However, since Volkersen didn't include any bending stresses, his model only applies when the Young's Modulus (stiffness) of the substrate (what the adhesive is bonded to) is much greater than the adhesive, so it applies for other kinds of joint configurations, not ours. Therefore, with ductile adhesives (like Loctite, since it can elongate up to 10%), use the following:

$$\tau = \frac{P}{bl} \tag{1}$$

Where b is the joint's width and l is the overlap length. Since we use carbon-fiber tubing, not flat plates as the previous equation assumes, we use the following:

$$\tau = \frac{P}{(\pi \times d) \times l} \tag{2}$$

Where d is the diameter at bonding. Goland came up with a better model compared to Volkersen's by accounting for bending moment:

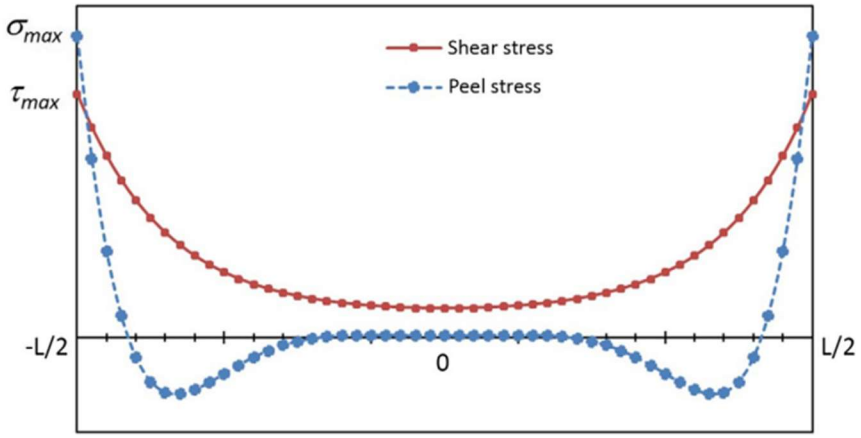
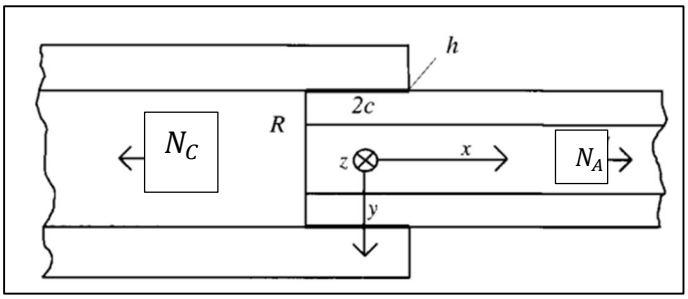


Fig. 15.3 Peel and shear stresses along an overlap in an SLJ using the Goland and Reissner approach.

We explore this relationship via the following equations.

On Shear Stress Distribution along Bond Length



$$\sum F_x = 0 = P - N_A(x) - N_C(x) \ - N_E(x) \tag{3}$$

Where x is in the axial direction and $x = 0$ at the middle-axial section. We cannot suppose axial deformation in the aluminum insert in the bond section at a distance x is equal to that of the carbon fiber tubing in the same region because the epoxy experiences shearing deformation. We set $N_E(x) = 0$ as the epoxy is thin and has a negligible axial reaction.

We note the equation for shear stress:

$$\tau_{rx} = -\frac{dN_C(x)}{2\pi R \cdot dx} \tag{4}$$

Where R is the radius at the bond interface at this point. Because this focuses on axial stress, the other two shear stress components of a typical stress element are negligible. We define strain in the axial direction as:

$$\gamma_{rx}(x) = \frac{\tau_{rx}(x)}{G_a} \tag{5}$$

Where G_a is the elastic shear modulus of the adhesive. As noted above in our first principal equation, the tensile force on the aluminum insert is distributed into 3 reactionary forces, where N_E is negligible. Therefore, the following relationship holds for some relationship $f(x)$:

$$N_C(x) = Pf(x), \quad N_A = P\big(1 - f(x)\big) \tag{6}$$

We set up our displacement equations as follows:

$$\sigma = E\varepsilon \rightarrow \delta = \frac{FL}{EA} \rightarrow \begin{cases} u_C = \int_{-c}^{x} \frac{N_C(x)}{E_C A_C} dx + u_C^0 \\ u_A = \int_{-c}^{x} \frac{N_A(x)}{E_A A_A} dx + u_A^0 \end{cases} \quad (7)$$

Where E_C and E_A are Young's modulus for carbon-fiber tubing and aluminum insert, respectively, followed by A_C & A_A for their respective cross-sectional areas, and u_C^0 & u_A^0 are displacements at the starting position, where u_A^0 can also be set to 0. To accommodate our insufficient number of equations, we need to set up the compatibility equation as follows:

$$\Delta u = u_2 - u_1 = h\gamma_{rx}(x) \quad (8)$$

Where h is the thickness of the epoxy. After plugging in Eq. 5 into Eq. 8, we introduce an adhesive stiffness per unit length, K^*:

$$K^* = \frac{2\pi R G_a}{h} \quad (10)$$

Plugging in Equation Set 7 into Eq. 8, and using the shared relationship between N_C and N_A, as well as normal boundary conditions, we eventually have a second-order differential equation for $f(x)$. For the purposes of this document, I will just list the final shear stress distribution. However, the full derivation can be found in this research document. The final shear stress is:

$$\tau_{rx}(x) = -\frac{N}{2\pi R}\frac{df(x)}{dx}$$

Generally, when the stiffnesses of the two tubes $(E_i A_i)$ are closer together, the stress peaks less at the ends and is more uniform, which is usually the better case. After the length of the bonded end is over, there is no more shear stress on the epoxy, of course. An important note is that this analysis does not include transverse shear deformation.

			Without Bump		With 3G Bump		
16	Tire moments (N*mm)	-Mx	Max Tension (N)	Max Compression (N)	Max Tension (N)	Max Compression (N)	
17		-My					
18		-Mz					
19							
20	Link Forces (N)	Upper Fore	320	-1647	-14	-2496	Upper Fore
21		Upper Aft	440	-2934	-91	-4541	Upper Aft
22		Lower Fore	3089	-750	4759	62	Lower Fore
23		Lower Aft	2669	-632	3847	-136	Lower Aft
24		Spring	-343	-3269	-1752	-7589	Spring
25		Tie Rod	1949	-270	3120	127	Tie Rod
26	Standstill Normal Forces (N)						

Using epoxy_stress_calculator.m code I created based on the equations above, Rockwest Composites Tubing 45511, and the above forces experienced by the suspension members:

Epoxy Shearing FoS: 3.611

Carbon-Fiber Tubing Compression FoS: 27.729

Euler Buckling on Carbon-Fiber Tubing FoS: 143.776

Therefore, we select Rockwest Composites Tubing 4551.